LTE Signaling

Course objectives:

The course offers a complete understanding of all signaling protocols implemented in the Long Term Evolution (LTE) system. Starting with an overview of the general architecture of the EPS system, it defines interfaces and protocol stacks, as well as presents the description of their functions and services. The course focuses on details of both the radio access network signaling and the core network signaling. Essential network procedures corresponding with the LTE protocols are discussed together with the relevant signaling flows. Examples of the discussed procedures are network attach, bearer establishment, UE context establishment, paging, X2/S1/UTRAN handover, random access, timing alignment, RLC retransmissions, HARQ etc. The course also explains the security aspects of the LTE/SAE including authentication, NAS and AS security. The course is based on the 3GPP Technical Specifications and hence is vendor-independent.

Duration Instructional methods:

Option 1: On-Site Training: 3 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 3 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 6 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to network engineers who will deal with LTE optimization, configuration, operation, maintenance or support, as well as any technical staff requiring deeper appreciation of LTE signaling.

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Course Contents:

- LTE Architecture and Components
  - Core network: SAE overview
  - Access Network: E-UTRAN overview
  - Roaming Architecture
  - Interworking with other Networks

- Generic LTE/SAE Protocol Architecture
  - User plane
  - Control Plane
  - Quality of Service Concept
  - EPS Bearers

- E-UTRAN Network Interface: S1
  - Protocol Structure
  - Initiation over S1
  - Context Management
  - Bearer Management
  - Paging over S1
  - Mobility over S1
  - Load management

- E-UTRAN Network Interface: X2
  - Protocol Structure
  - Initiation over X2
  - Mobility over S2
  - Load and Interference Management
  - UE Historical Information over X2

- Control Plane Protocols: Radio Resource Control - RRC
  - RRC states
  - RRC procedure and messages
  - System Information
  - Connection Control in LTE
  - Connected Mode Inter-RAT Mobility
  - Measurements
  - PLMN and Cell Selection
  - Cell Reselection
  - Paging

- User Plane Protocols: Packet Data Convergence Protocol - PDCP
  - Functions and Architecture
  - Header Compression
  - Security
  - Handover
  - PDU Formats
  - Discard of Data Packets

- User Plane Protocols: Radio Link Control - RLC
  - RLC Entities
    - Transparent Mode - TM -Entity
    - Unacknowledged Mode - UM - Entity
    - Acknowledged Mode - AM - Entity
  - RLC PDU Formats

- User Plane Protocols: Medium Access Control - MAC
  - MAC Architecture
  - Mapping of Logical Channels to Transport Channels
  - MAC Procedures
  - Random Access
  - Uplink Time Alignment
  - Downlink Data Transfer
  - Uplink Data Transfer
  - PDUs and Formats

- Non-Access Stratum - NAS Signaling
  - NAS Protocol States and Transitions
  - NAS Security
    - EPS key hierarchy
    - AKA (Authentication and Key Agreement)
    - Authentication procedure
    - NAS signaling protection
    - AS signaling and user data protection
  - Non-Access Stratum Protocols
  - Evolved Mobility Management - EMM
  - Evolved Session Management - ESM
  - Mobility Management across EMM States
  - EMM Procedures

- Evolved GTP: GTPv2-C
  - GTP Stack
  - GTP Format
  - Messages
  - Path Management
  - Tunnel Management
  - Mobility Management

Pre-requisites:

Training participants should bring a deep understanding of the EPC architecture, nodes and functions.
Alternatively, prior to this training, participants should attend the course LTESysO.

Course Code: LTESig

Price per student:

For Option 1, Classroom training : USD 1,800.00 per student

For Options 2, 3, Virtual Classroom training: USD 1,500.00 per student